

HealthServlet.java

```
package com.harisudhan.harisudhanmart.controller;

import com.harisudhan.harisudhanmart.listener.AppContextListener;
import java.io.IOException;
import java.sql.Connection;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

/** Section 18: GET /api/v1/health returns { "status": "UP", "db": "UP" }, verifying DB connectivity. */
@WebServlet("/api/v1/health")
public class HealthServlet extends BaseServlet {

    @Override
    protected void doGet(HttpServletRequest req, HttpServletResponse resp) throws ServletException, IOException {
        String dbStatus;
        try (Connection conn = AppContextListener.getDataSource().getConnection()) {
            dbStatus = conn.isValid(2) ? "UP" : "DOWN";
        } catch (Exception e) {
            dbStatus = "DOWN";
        }
        resp.setContentType("application/json");
        resp.setStatus(HttpServletResponse.SC_OK);
        resp.getWriter().write(String.format("{\"status\":\"UP\", \"db\":\"%s\"}", dbStatus));
    }
}
```